

4NF AND 5NF

Prof. Geetha Mary A
Professor,
SCOPE, VIT, Vellore

ALGORITHMS FOR RELATIONAL DATABASE SCHEMA DESIGN (5)

ISSUES WITH NULL-VALUE JOINS. (A) SOME EMPLOYEE TUPLES HAVE NULL FOR THE JOIN ATTRIBUTE DNUM.

(a)

EMPLOYEE

ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUM
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4
Narayan, Ramesh K.	666884444	1962-09-15	975 Fire Oak, Humble, TX	5
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1
Berger, Anders C.	999775555	1965-04-26	6530 Braes, Bellaire, TX	null
Benitez, Carlos M.	888664444	1963-01-09	7654 Beech, Houston, TX	null

DEPARTMENT

DNAME	<u>DNUM</u>	DMGRSSN
Research	5	333445555
Administration	4	987654321
Headquarters	1	888665555

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ISSUES WITH NULL-VALUE JOINS. (B) RESULT OF APPLYING NATURAL JOIN TO THE EMPLOYEE AND DEPARTMENT RELATIONS. (C) RESULT OF APPLYING LEFT OUTER JOIN TO EMPLOYEE AND DEPARTMENT.

(b)

ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUM	DNAME	DMGRSSN
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 Fire Oak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

(c)

ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUM	DNAME	DMGRSSN
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 Fire Oak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555
Berger, Anders C.	999775555	1965-04-26	6530 Braes, Bellaire, TX	null	null	null
Benitez, Carlos M.	888664444	1963-01-09	7654 Beech, Houston, TX	null	null	null

THE “DANGLING TUPLE” PROBLEM. (B) THE RELATION EMPLOYEE_2 (INCLUDES DNUM ATTRIBUTE WITH NULL VALUES). (C) THE RELATION EMPLOYEE_3 (INCLUDES DNUM ATTRIBUTE BUT DOES NOT INCLUDE TUPLES FOR WHICH DNUM HAS NULL VALUES).

(b) **EMPLOYEE_2**

<u>SSN</u>	DNUM
123456789	5
333445555	5
999887777	4
987654321	4
666884444	5
453453453	5
987987987	4
888665555	1
999775555	null
888664444	null

(c) **EMPLOYEE_3**

<u>SSN</u>	DNUM
123456789	5
333445555	5
999887777	4
987654321	4
666884444	5
453453453	5
987987987	4
888665555	1



MULTIVALUED DEPENDENCIES AND FOURTH NORMAL FORM (1)

(A) THE EMP RELATION WITH TWO MVDs: $\text{ENAME} \twoheadrightarrow \text{PNAME}$ AND $\text{ENAME} \twoheadrightarrow \text{DNAME}$. (B) DECOMPOSING THE EMP RELATION INTO TWO 4NF RELATIONS EMP_PROJECTS AND EMP_DEPENDENTS.

(a)

EMP

<u>ENAME</u>	PNAME	<u>DNAME</u>
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Smith	X	John
Smith	Y	Anna
Smith	X	Anna
Smith	Y	John

(b)

EMP_PROJECTS

<u>ENAME</u>	<u>PNAME</u>
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Smith	X
Smith	Y

EMP_DEPENDENTS

<u>ENAME</u>	<u>DNAME</u>
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Smith	John
Smith	Anna



MULTIVALUED DEPENDENCIES AND FOURTH NORMAL FORM (2)

Definition:

- A **multivalued dependency (MVD)** $X \twoheadrightarrow Y$ specified on relation schema R , where X and Y are both subsets of R , specifies the following constraint on any relation state r of R : If two tuples t_1 and t_2 exist in r such that $t_1[X] = t_2[X]$, then two tuples t_3 and t_4 should also exist in r with the following properties, where we use Z to denote $(R - (X \cup Y))$:
 - $t_3[X] = t_4[X] = t_1[X] = t_2[X]$.
 - $t_3[Y] = t_1[Y]$ and $t_4[Y] = t_2[Y]$.
 - $t_3[Z] = t_2[Z]$ and $t_4[Z] = t_1[Z]$.
- An MVD $X \twoheadrightarrow Y$ in R is called a **trivial MVD** if (a) Y is a subset of X , or (b) $X \cup Y = R$.

- **Definition.** A relation schema R is in 4NF with respect to a set of dependencies F (that includes functional dependencies and multivalued dependencies) if, for every *nontrivial* multivalued dependency $X \twoheadrightarrow Y$ in F^+ , X is a superkey for R .



4. JOIN DEPENDENCIES AND FIFTH NORMAL FORM (1)

Definition:

- A **join dependency (JD)**, denoted by $JD(R_1, R_2, \dots, R_n)$, specified on relation schema R , specifies a constraint on the states r of R . The constraint states that every legal state r of R should have a non-additive join decomposition into $R_1, R_2, \dots, R_n$; that is, for every such r we have

$$* (\pi_{R_1}(r), \pi_{R_2}(r), \dots, \pi_{R_n}(r)) = r$$

***Note:** an MVD is a special case of a JD where $n = 2$.*

- A join dependency $JD(R_1, R_2, \dots, R_n)$, specified on relation schema R , is a **trivial JD** if one of the relation schemas R_i in $JD(R_1, R_2, \dots, R_n)$ is equal to R .

JOIN DEPENDENCIES AND FIFTH NORMAL FORM (2)

Definition:

- A relation schema R is in **fifth normal form (5NF)** (or **Project-Join Normal Form (PJNF)**) with respect to a set F of functional, multivalued, and join dependencies if, for every nontrivial join dependency $JD(R_1, R_2, \dots, R_n)$ in F^+ (that is, implied by F), every R_i is a superkey of R .

RELATION SUPPLY WITH JOIN DEPENDENCY AND CONVERSION TO FIFTH NORMAL FORM

(C) THE RELATION SUPPLY WITH NO MVDs IS IN 4NF BUT NOT IN 5NF IF IT HAS THE JD(R1, R2, R3).

(D) DECOMPOSING THE RELATION SUPPLY INTO THE 5NF RELATIONS R1, R2, AND R3.

(c) **SUPPLY**

SNAME	PARTNAME	PROJNAME
Smith	Bolt	ProjX
Smith	Nut	ProjY
Adamsky	Bolt	ProjY
Walton	Nut	ProjZ
Adamsky	Nail	ProjX
Adamsky	Bolt	ProjX
Smith	Bolt	ProjY

(d) **R1**

SNAME	PARTNAME
Smith	Bolt
Smith	Nut
Adamsky	Bolt
Walton	Nut
Adamsky	Nail

R2

SNAME	PROJNAME
Smith	ProjX
Smith	ProjY
Adamsky	ProjY
Walton	ProjZ
Adamsky	ProjX

R3

PARTNAME	PROJNAME
Bolt	ProjX
Nut	ProjY
Bolt	ProjY
Nut	ProjZ
Nail	ProjX